

Hardware-Based Detection and Prevention of Ultrasonic Voice Command Attacks (DolphinAttack)

Ben Israel Gurjar

Yeshua Macwan

Luis Angel Ochoa

Aryan Singh

College of Engineering & Computer Science, CSU Sacramento, Sacramento California, USA

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Abstract: This project presents a hardware-based ultrasonic attack detection and prevention system designed to protect voice-controlled devices from DolphinAttack-style ultrasonic voice command attacks. DolphinAttack exploits the nonlinear behavior of microphones by transmitting hidden voice commands at ultrasonic frequencies above the range of human hearing, allowing devices such as Siri, Alexa, and Google Assistant to unintentionally process malicious commands. The proposed system uses an ultrasonic microphone, a high-pass filter, a two-stage operational amplifier, a full-wave bridge rectifier, a NAND gate logic stage, LED indicators, and a transistor switching stage to detect ultrasonic activity and disable the normal microphone path when an attack is present. Experimental testing showed that the ultrasonic microphone successfully detected high-frequency signals, but the amplifier and rectifier stages did not consistently produce enough DC voltage to fully activate the LED indicators or switching stage due to the extremely small microphone signal amplitude, diode voltage losses, and circuit noise. Despite these limitations, the project demonstrated the feasibility of using analog hardware circuitry for ultrasonic attack detection and provided valuable insight into ultrasonic sensing, amplification, filtering, and hardware-based security protection methods.

Introduction: Fundamentally, the ultrasonic attack detection system is a hardware-based protection circuit designed to identify inaudible ultrasonic signals and prevent them from being processed by a normal microphone path. This project was developed in response to a security vulnerability known as *DolphinAttack* [1], where voice-controlled systems such as Siri, Alexa, and Google Assistant can be manipulated using ultrasonic voice commands above 20 kHz. These frequencies are outside the range of human hearing, but microphones can still detect them. Due to the nonlinear behavior of microphone hardware, the ultrasonic signal can be unintentionally demodulated into an audible command that a speech recognition system can process and execute. As a result, an attacker may be able to issue malicious commands to a device without the user hearing or noticing the attack.

The purpose of this project is to detect ultrasonic activity before it is processed by the speech recognition system and to disable the microphone path when a potential attack is present. Omitting basic components such as wires, resistors, capacitors, and power connections, the main components of the detection circuit are as follows: (a) an ultrasonic microphone used to sense incoming high-frequency signals, (b) a high-pass filter used to pass ultrasonic frequencies, (c) an operational amplifier used to increase the amplitude of the detected signal, (d) a full-wave bridge rectifier and capacitor filter used to convert the AC signal into a DC control signal, (e) a NAND gate configured as an inverter, (f) a green and red light emitting diode (LED) used as status indicators, and (g) a transistor switching stage used to disable the normal microphone path.

Based on the circuit design, when no ultrasonic signal is detected, the normal microphone path remains active, and the system allows regular voice input to pass through. When an ultrasonic signal greater than approximately 20 kHz is detected, the signal is first amplified by the op-amp and then passed through the high-pass filter. The filtered AC signal is then converted into a DC voltage by the full-wave rectifier and capacitor filter. This DC control signal is applied to the logic stage, where the NAND gate changes state and drives the transistor switch that once triggered, the transistor disables the normal microphone path, preventing the ultrasonic attack signal from being processed as a valid voice command. A red LED is used to indicate when ultrasonic activity has been detected, while the normal microphone output is blocked during the protection state, and green when no attack is present, allowing for normal usage from the user. A picture of the working prototype can be seen in Figure 1.

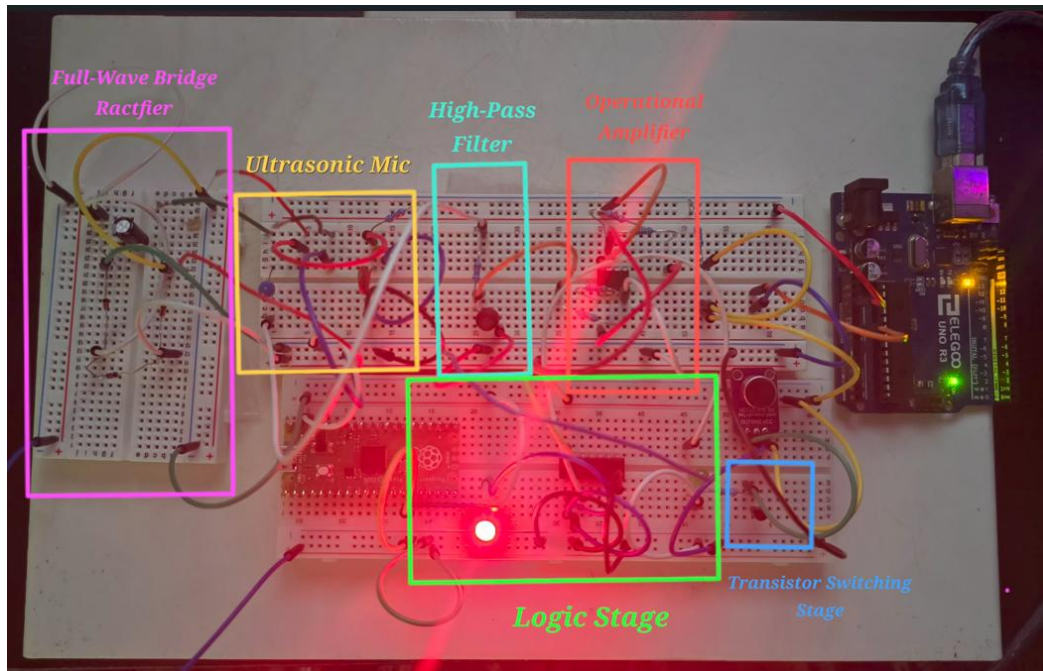


Figure 1: Hardware-Based Detection and Prevention of Ultrasonic Voice Command Attack Circuit.

Technical details: The circuit diagram of the ultrasonic attack detection and microphone disable system is shown in Figure 2. Each stage of the device is designed to detect ultrasonic activity and disable the microphone path when a potential attack is present. The main components of the system are discussed below.

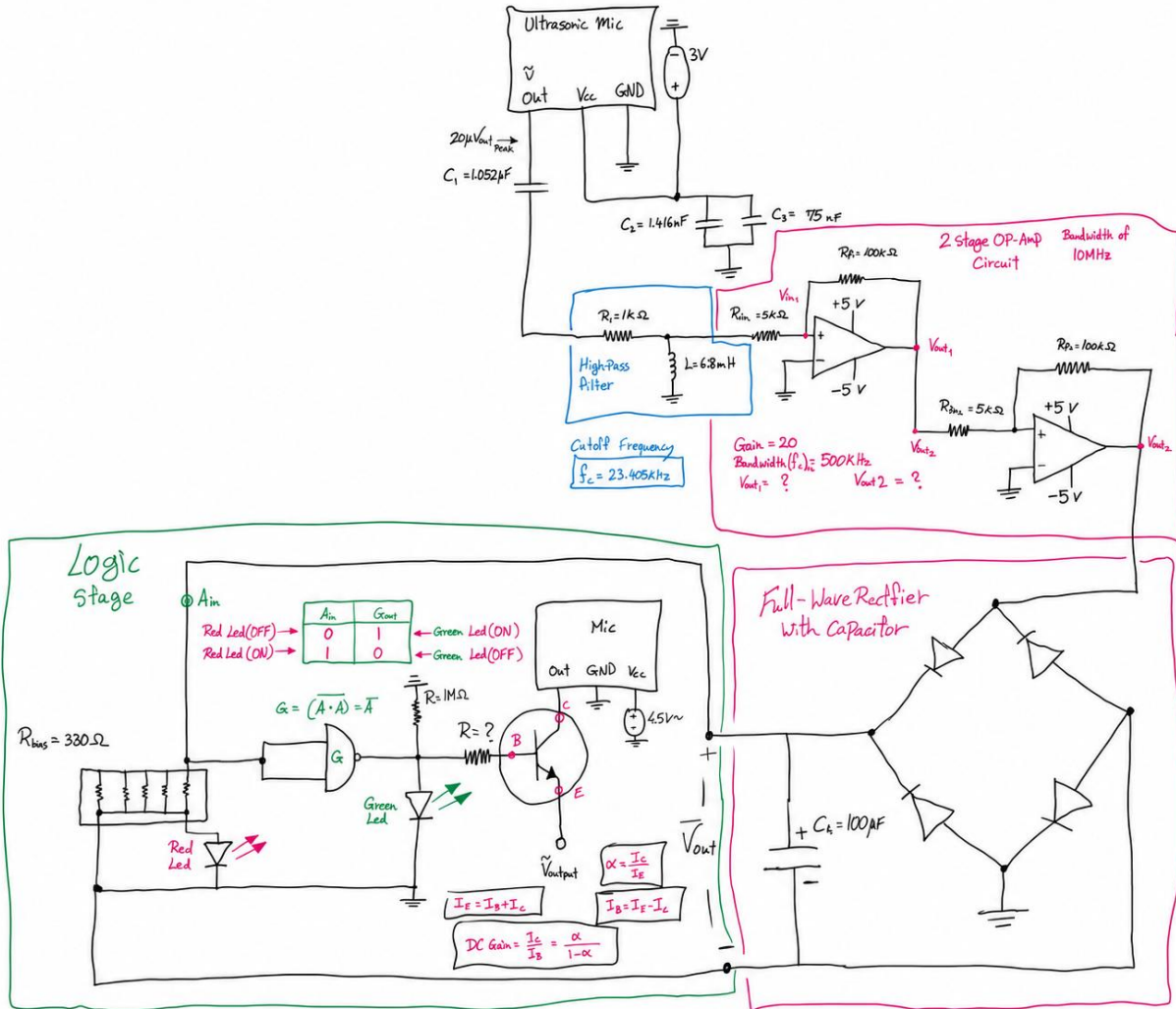


Figure 2: Hardware-Based Detection and Prevention of Ultrasonic Voice Command Attack Schematic.

a) Ultrasonic Microphone Input Stage: The ultrasonic microphone is used as the sensing element for the detection circuit. It is powered by using a 3 V supply and is responsible for converting incoming ultrasonic acoustic signals into a small electrical signal. Microphones, including MEMS and electret types, work by converting acoustic pressure variations into corresponding electrical signals, enabling detection of both audible and ultrasonic frequencies depending on their design characteristics [2], [3], [4].

Since DolphinAttack relies on frequencies above 20 kHz [1], the purpose of this microphone is to detect high-frequency sound that is inaudible to humans but still present in the environment. Under the test conditions, the microphone output was measured to be approximately $20\mu V_{pk-pk}$, which is a very small signal and because of this, the output of the microphone must be filtered and amplified before it can be used by the logic stage.

Capacitors placed near the microphone are used for both power supply decoupling and signal coupling. A local bypass capacitor helps reduce noise on the supply line, while coupling capacitors are used to block DC components and stabilize the signal before it enters the next stage of the circuit [3], [5], [6].

b) High-pass Filter Stage: The signal from the ultrasonic microphone is passed through a high-pass filter [7]. The purpose of this stage is to weaken lower-frequency signals, including normal audible voice signals, while allowing ultrasonic frequencies to pass. This is important because the detection circuit should respond primarily to ultrasonic activity rather than ordinary speech or background noise so that no accidental trigger of the switch may occur during normal use. The high-pass filter is implemented using an RL configuration with component values:

$$R = 1\text{ k}\Omega, L = 6.8\text{ mH}$$

The cutoff frequency for an RL high-pass filter is given by:

$$f_c = \frac{R}{2\pi L} \quad (1)$$

Substituting the component values,

$$f_c = \frac{1000}{2\pi(6.8 \times 10^{-3})}$$

$$f_c \approx 23.405 \text{ kHz}$$

This cutoff frequency is above the upper limit of human hearing and lies within the ultrasonic range used in DolphinAttack [1]. Therefore, the filter suppresses lower-frequency audio signals while allowing ultrasonic components to pass through the amplification stage.

c) Two-Stage Operational Amplifier Circuit: After filtering, the signal is sent to a two-stage operational amplifier circuit that is used to increase the amplitude of the weak ultrasonic microphone signal so that it can be processed by the rectifier and logic stages. Based on the schematic, each amplifier stage uses the following component values:

$$R_f = 100 \text{ k}\Omega, R_{in} = 5 \text{ k}\Omega$$

If the stage is configured as an inverting amplifier, the voltage gain is given by:

$$\text{Gain}(A_v) = -\frac{R_f}{R_{in}} \quad (2)$$

Substituting the component values,

$$\text{Gain}(A_v) = -\frac{100 \text{ k}\Omega}{5 \text{ k}\Omega} = -20$$

Since there are two amplifier stages, the total gain is approximately,

$$\text{Total Gain}(A_{v \text{ total}}) = A_{v1} \times A_{v2} \quad (3)$$

Since both amplifier stages use identical component values, the total gain is

$$Total\ Gain(A_{v\ total}) = (-20) \times (-20) = 400$$

The second stage reverses the inversion from the first stage, resulting in a final output signal that is amplified by approximately 400 times compared to the input.

The amplifier used in this design is the MCP6022, a dual operational amplifier integrated circuit that contains two independent op-amps on a single chip. The MCP6022 operates using a single supply voltage in the range of 2.5 V to 5.5 V and has a typical gain bandwidth product (GBW) of 10 MHz and a slew rate of approximately $7\text{ V}/\mu\text{s}$ [8].

The closed loop bandwidth of each amplifier stage can be estimated using

$$f_B = \frac{GBW}{|A_v|} \quad (4)$$

Substituting the given values,

$$f_B = \frac{10\text{ MHz}}{20} = 500\text{ kHz}$$

Since 500 kHz is significantly greater than the high-pass filter cutoff frequency of approximately 23.405 kHz, the amplifier can amplify ultrasonic signals within the desired frequency range without significant loss due to bandwidth limitations.

Additionally, the slew rate of $7\text{ V}/\mu\text{s}$ is sufficient to support the rate of change of the amplified ultrasonic signal, ensuring that the output waveform can follow the rapid changes of the ultrasonic signal without significant distortion [8].

d) Full-Wave Bridge Rectifier (AC-DC Converter): The output of the two-stage operational amplifier is an amplified AC signal. For the logic stage to detect whether ultrasonic activity is present, this AC signal must be converted into a DC control signal. This is done by using a full-wave bridge rectifier followed by a smoothing capacitor [9].

A full-wave rectifier uses four diodes arranged in a bridge configuration to convert both the positive and negative halves of the input AC waveform into a pulsating DC signal [9].

This ensures that the output maintains a consistent signal regardless of the input signal direction.

The output after the rectifier is then passed through a capacitor filter, which smooths the pulsating DC signal by storing and releasing charge, which results in a more stable DC voltage that can be used by the logic stage for more consistent switching. The capacitor used in this design is:

$$C_h = 100 \mu F$$

The approximate DC output voltage of a full-wave bridge rectifier is given by

$$V_{DC} \approx V_{peak} - 2V_D \quad (5)$$

where V_{peak} is the peak value of the amplified input signal and V_D is the forward voltage drop of a single diode. Since two diodes conduct each half-cycle in a bridge rectifier, the voltage drop is approximately twice the diode forward voltage [9].

Since the NAND gate logic circuit cannot reliably process a small AC signal directly, this stage of the circuit is critical. By converting the ultrasonic signal into a DC voltage level, the rectifier and capacitor filter allow the system to determine whether ultrasonic activity is present.

e) NAND Gate Logic Stage: The DC control signal from the rectifier is applied to a NAND gate configured as an inverter by tying both inputs together, producing the logical complement of the input.

The Boolean expression for a NAND gate is given by

$$G = \overline{A \cdot A} \quad (6)$$

Since

$$\overline{A \cdot A} = \overline{A}$$

the expression simplifies to

$$G = \overline{A}$$

This means that the NAND gate inverts the input signal. When no ultrasonic signal is detected, the rectifier output remains low, resulting in a high output from the NAND gate. When ultrasonic activity is detected, the rectifier output becomes high, causing the NAND gate output to switch to a low state.

This is used to control both the LED indicator stage and the transistor switching stage. The change in output state ensures that the system can distinguish between normal operation and the presence of an ultrasonic attack signal.

The truth table for the NAND gate configured as an inverter is shown in Table 1.

Input A_{in}	Output G_{out}	System State
0	1	Normal Operation – Green LED ON
1	0	Ultrasonic signal detected – Red LED ON

Table 1. Truth table for NAND gate configured as an inverter.

f) LED Indicator Stage: Light emitting diodes (LEDs) are used in the circuit to provide a visual indication of the system state. A green LED is used to indicate normal operation, while a red LED indicates the detection of ultrasonic activity.

The LEDs are controlled by the output of the NAND gate. When no ultrasonic signal is detected, the NAND gate output remains high, turning on the green LED and indicating normal operation. When an ultrasonic signal is detected, the NAND gate output transitions to a low state, turning on the red LED and indicating that the system has entered its protection mode.

g) Transistor Switching Stage: The final control stage of the circuit uses a bipolar junction transistor (BJT) as an electronic switch to disable the normal microphone path when ultrasonic activity is detected. The transistor is controlled by the output of the NAND gate.

When no ultrasonic signal is present, the NAND gate output remains high and the circuit remains in its normal operating state. In this condition, the normal microphone path remains active, allowing regular voice input to pass through the system. When ultrasonic activity is detected, the rectifier output changes the NAND gate input state, causing the NAND gate output to transition low. This logic transition drives the transistor switching

stage and disables the normal microphone path, preventing the ultrasonic signal from reaching the speech recognition system.

For a BJT, the collector current is controlled by the base current, allowing a small control signal at the base to switch a larger current path at the collector-emitter terminals [11].

The transistor current relationship is given by

$$I_E = I_B + I_C \quad (7)$$

where I_E is the emitter current, I_B is the base current, and I_C is the collector current. The DC current gain of the transistor is expressed as

$$DC\ Gain(\beta) = \frac{I_C}{I_B} \quad (8)$$

This relationship shows that a small base current can control a larger collector current [10], allowing the transistor to function as a switch. In this project, the switching stage is critical because it provides the final protective action: disabling the microphone path whenever ultrasonic activity is detected.

Experimental Results and Discussion:

During testing, the AC-to-DC rectifier output was only in the millivolt range, which was not sufficient to directly turn on the LED indicator. This occurred because the microphone output signal was very small, and the diode voltage drops in the rectifier reduced the available output voltage. A future improvement would be to use the rectified voltage only as a control signal rather than as the LED power source. The circuit could be redesigned with a comparator stage followed by a transistor or MOSFET switch, allowing a small detected ultrasonic voltage to trigger an LED powered by a separate 5 V supply. In addition, Schottky diodes or an op-amp precision rectifier could be used to reduce voltage loss and improve sensitivity.

The second stage did not perform as intended because the signal became heavily attenuated after passing through the high-pass filter and amplification stage. Although the first stage produced an ultrasonic signal of about 215.625 mV peak-to-peak, the output of the second stage dropped to

only a few millivolts, which made the response to changes in sound level and frequency very difficult to observe. This suggests that the second stage was either filtering out too much of the desired signal or not providing enough usable gain after filtering. In addition, the MCP6022 op-amp did not amplify the ultrasonic signal cleanly; instead, background noise was amplified along with the signal, which reduced the signal-to-noise ratio. Because the noise became comparable to or larger than the desired signal, the detector's stage was triggered unpredictably, causing false alarms and random activation of the red LED. Overall, the failure of the second stage was mainly caused by excessive attenuation of the ultrasonic signal and noisy amplification.



Figure 2: Oscilloscope measurement of the first-stage ultrasonic output signal, showing a peak-to-peak voltage of approximately 215.625 mV.



Figure 3: Oscilloscope measurement of the second-stage output after the high-pass filter, showing significant attenuation of the signal.

Future Work:

Future work will focus on reducing distortion in the regular microphone path and improving the signal-to-noise ratio of the ultrasonic detection stage. The op-amp, high-pass filter, rectifier, and transistor stages should be further optimized so that the circuit can detect ultrasonic attacks more reliably without false triggering. The complete system will also need to be integrated and tested as one circuit, followed by additional measurements at different ultrasonic frequencies and amplitudes to improve performance and stability.

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